

LLM-Guided Synthesis of Constraint-Adaptive Heuristics for Satellite–Ground Scheduling

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Abstract

Graph optimization heuristics can achieve strong aggregate performance while relying on decision rules that fail under changes in resource constraints. Effective adaptation requires identifying not only when action preferences should change, but also whether the input representation contains the information needed for that change. We propose an LLM-guided synthesis framework for constraint-adaptive satellite–ground scheduling that uses certified decision evidence to jointly refine graph representations and ranking rules. Its central mechanism detects ranking requirements that become contradictory under the current feature representation and directs synthesis toward the missing structural distinctions, rather than repeatedly adjusting an insufficient scoring interface. Two complementary mechanisms support this refinement. First, controlled offline interventions on resource-constraint configurations generate paired static scheduling problems in which the same competing actions remain individually feasible. Sound bounds on conditional completion values certify preference reversals or preservation under explicit boundary commitments, yielding relational specifications of when rankings should change or remain stable. Second, structural witnesses guide the LLM to compose distinguishing features from a typed graph-operation library and synthesize corresponding ranking rules. Feature–rule pairs are selected jointly for declared-interface acyclicity, complete-schedule quality, and feature-computation cost, making representation expansion targeted and cost-aware. We formulate contact selection as maximum-weight independent set optimization and study adaptation using a frozen program on held-out scheduling instances and constraint configurations. The synthesized program responds to the current conflict graph within a fixed feasibility-enforcing kernel, without online LLM or oracle calls. By linking representation refinement to certified decision conflicts, the framework addresses information limitations beyond rule-only repair while explicitly accounting for the computational cost of constraint adaptation.

CCS Concepts

• **Computing methodologies** → **Planning and scheduling**; • **Mathematics of computing** → *Combinatorial optimization*.

Keywords

Satellite scheduling, maximum-weight independent set, heuristic synthesis, representation refinement, large language models

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1 Introduction

A satellite–ground contact commits scarce resources and excludes competing opportunities. Their forgone reward depends on whether they could coexist. Resource thresholds change this relationship while contact windows and rewards remain fixed. Operational scheduling models coupled resources [4, 22, 28]; we study pairwise-conflict scheduling, where a reusable maximum-weight independent set (MWIS) heuristic must respond to different static configurations.

LLMs synthesize executable heuristics using objective feedback in FunSearch, EoH and ReEvo [25, 29, 36]. DRAGON uses decomposition/reconstruction agents with boundary constraints and checker feedback [8]. We ask how synthesis can identify missing inputs that make required rankings impossible: formula revision cannot satisfy incompatible strict preferences over identical observations.

Figure 1 constructs this obstruction. Actions a, b have equal rewards and base features. Their forced-completion increments are 20, 26 before a ground-gap intervention and 20, 14 afterward. Both remain feasible; the preference reverses, but every pointwise base scorer ties them within each configuration. Shared resource fields cannot distinguish them; neighborhood structure can. These values concern forced completions, not an unrestricted rollout’s reward.

We propose certified representation–rule refinement. Following abstraction refinement and planning-feature precedents [9, 14], we project sound strict preferences onto exact feature-equivalence classes. Directed cycles certify an information obstruction. Witnesses retain ranking arcs and equality joins, targeting LLM proposals toward missing structural distinctions. Full refined quotients are repeatedly checked: separating one witness can leave other cycles intact.

Shared residual components cancel from value differences without bounding their unknown optima. Reached argmax audits retain finite-pool regret bounds and unknowns. Demand-driven execution shares typed computations and charges work. TRAIN selection balances complete-schedule reward and cost, with reached regret ranking within a declared utility band. Our contributions are:

- (1) **Certified information diagnosis**: finite pointwise representability characterized by quotients and concrete cycle witnesses.
- (2) **Witness-guided repair**: full-quotient separation and minimum additive-cost refinement over fixed evidence and a bounded finite catalogue.



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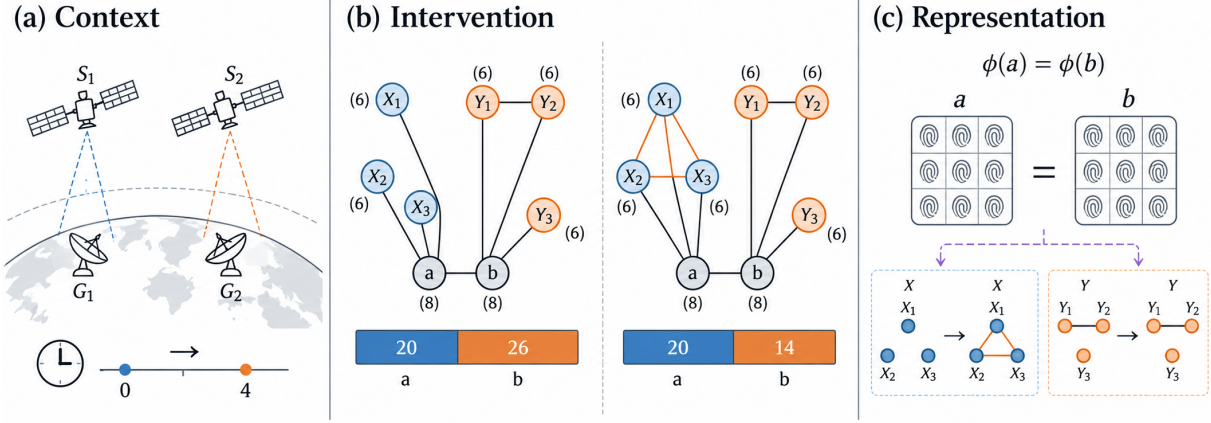


Figure 1: Why a ranking rule can need new inputs. A constructed ground-gap intervention changes neighborhood compatibility and reverses forced-completion preferences, while preserving the nine-input alias within each configuration. Neighborhood structure supplies the missing distinction. Values exclude the same fixed boundary reward; the physical scene illustrates the domain.

- (3) **Decision-aware synthesis:** cancelled evidence, reached-action audits and demanded computation support joint selection.

Frozen scheduling and public-graph confirmation compare the selected program with published MWIS implementations. A separate four-block, 144-proposal cold-authoring study contrasts explicit witnesses, certified relations and objective feedback on new held-out configurations. TRAIN-only selection precedes all confirmations. A fixed kernel enforces graph feasibility without online LLM or oracle calls; quality and transfer remain empirical.

2 Related Work

Scheduling and graph decisions. Mixed-integer decomposition [4], integrated collection/downlink planning [22], and ground-queue-aware traffic scheduling [31] establish resource context’s importance. Our pairwise model excludes arbitrary cumulative constraints. Graph embeddings condition optimization on partial solutions [10]; learned deferral changes independent-set decision stages [1]. AAMAS work combines factor-graph inference with routing [23] and local topology with neural representations [5]. Structural inputs are established choices. Published weighted branch-and-reduce [21], Struction [15], memetic kernelization [17] and concurrent local search [18] provide our specialized comparators. Exchange neighborhoods [3] and coloring-based clique search [27] are classical building blocks; our weighted exchange control is not an ARW reproduction.

GNN expressiveness analyses characterize aggregation indistinguishability [26, 35]. Our exact declared-feature quotient instead certifies an obstruction for every pointwise scalar scorer on observed occurrences; it is neither graph isomorphism nor a bound on all policies. Structural witnesses also appear in AAMAS universal motion-reachability theory [11], addressing a different guarantee. Dynamic heuristic selection uses search state [30], while algorithm configuration optimizes over instance distributions and controls

evaluation effort [20]. Adaptivity and cost accounting alone are therefore not novel; we diagnose the distinctions a scorer must see.

LLM algorithm discovery. FunSearch searches priorities inside fixed skeletons [29]; EoH evolves thoughts/code [25]; ReEvo uses reflection [36]; LLaMEA evolves executable metaheuristics through fitness feedback [32]. EoH-S optimizes complementary heuristic sets [24]; LACE combines verified interfaces, tools and runtime-constrained portfolio evolution [16]. Offline discovery and cost budgets are precedents. DRAGON’s per-instance agents transmit boundary constraints and checker feedback [8]. Our distinction is diagnosing whether an interface can express certified conditional rankings and targeting its missing structure. Zero online calls alone establish no advantage.

Abstraction and synthesis. Counterexample-guided refinement uses failure witnesses [9]. Syntax-guided synthesis [2] separates candidate syntax from behavioral specifications. Abstraction-guided synthesis refines program semantics [33]; relational synthesis constrains relationships among calls or programs [34]. Generalized planning learns symbolic features/actions [6], including feature selection and transition separation [14], and changes feature/action representations [7]. Exact game-strategy synthesis separates state classification from executable action conditions [13]. DreamCoder learns reusable components with synthesis guidance [12]. We couple aligned resource interventions and feasible boundary replay to sound rankings, equality-join feature selection and repeated full-quotient separation. Acyclicity removes observed information obstructions; bounded-rule fit, shared cost and schedule quality remain distinct requirements.

3 Problem Setting

Each contact $v \in V$ specifies satellite σ_v , ground station γ_v , fixed interval $[s_v, e_v]$ and nonnegative reward w_v . Selection reserves the full interval. Source rewards are duration $e_v - s_v$; synthetic rewards

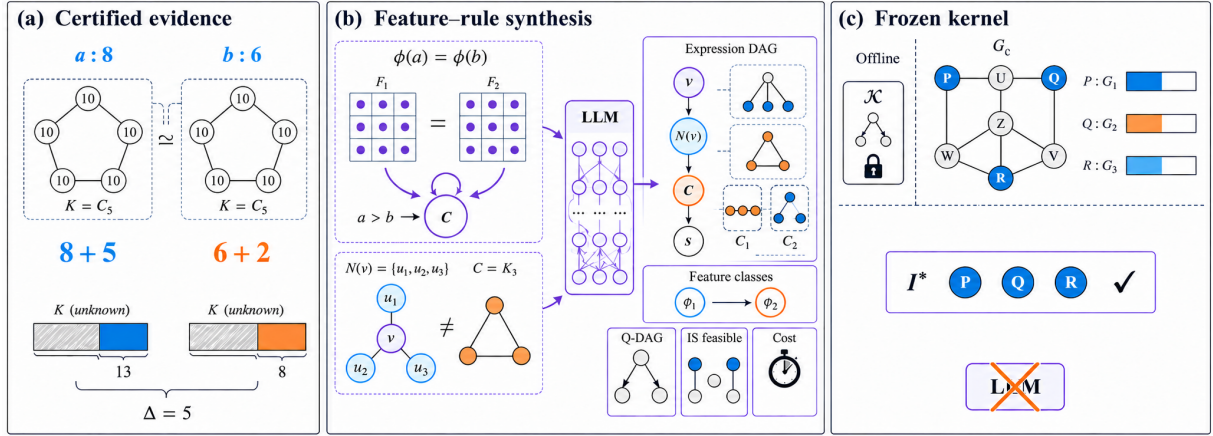


Figure 2: From certified conflicts to an executable heuristic. Common K cancels: $\Delta = (8 + 5) - (6 + 2) = 5$. **A separate equal-reward witness guides offline typed synthesis. Quotient, schedule and cost checks precede freezing $w/\max(10^{-6}, w, n_c)$, where n_c is neighbor clique-cover weight. The unit-weight toy returns optimum $\{P, Q, R\}$; bars identify contacts rather than physical conflict provenance. Deployment uses no LLM or oracle.**

may differ. Configuration θ fixes pairwise compatibility throughout a run (Figure 1).

Graph construction is model-specific. For contacts u, v ordered by start time, a shared ground station creates a conflict when $s_v < e_u + \tau_g$, where $\tau_g \geq 0$ is the station transition time. Shared-satellite conflicts depend on nonnegative change and transition thresholds τ_c, τ_t and the interval relation. Source-derived instances retain their declared satellite predicate, including its admissible crossing overlaps.¹ Synthetic probes use a separately declared single-capacity overlap model. Interventions preserve the declared model.

The graph $G_\theta = (V, E_\theta, w)$ joins pairs rejected by these predicates. With $w(I) = \sum_{v \in I} w_v$, contact selection is MWIS:

$$\max_{I \subseteq V} w(I). \quad (1)$$

I independent in G_θ

Feasibility requires all evaluated hard constraints to be pairwise encoded.

A boundary $B = (F, X)$ records previously selected contacts F and explicit exclusions X . It is admissible when F is independent, $F, X \subseteq V$, and $F \cap X = \emptyset$. Let $N_\theta(F) = \bigcup_{v \in F} N_\theta(v)$, where $N_\theta(v)$ is the neighbor set of v . The available actions are

$$R_\theta(B) = V \setminus (F \cup X \cup N_\theta(F)). \quad (2)$$

For a feasible next action $d \in R_\theta(B)$, define its conditional completion value as

$$V_\theta(d | B) = \max_{\substack{I \text{ independent in } G_\theta \\ F \cup \{d\} \subseteq I, I \cap X = \emptyset}} w(I). \quad (3)$$

This includes fixed reward and globally compatible future contacts.

Aligned configurations θ_0, θ_1 change one resource parameter, sharing contacts, intervals, identities, rewards and an admissible

boundary. Competing a, b are available and adjacent on both sides; other changed conflicts can alter their completion values.

A frozen $P = (\phi, h)$ assigns a finite score to every available contact, $h(\phi(G_\theta, R, v))$. The kernel commits an argmax with deterministic ties, deletes its closed neighborhood and repeats until $R = \emptyset$. This guarantees a complete feasible selection. Adaptation uses current graph and permitted resource fields, without online LLM or oracle calls.

4 Certified Representation–Rule Refinement

Certified conflicts direct offline typed feature–rule synthesis; representability, schedule quality and cost checks precede freezing (Figure 3).

4.1 Component-Cancelled Full-Residual Certificates

At boundary $B = (F, X)$, let $R = R_\theta(B)$, $R_d = R \setminus (\{d\} \cup N_\theta(d))$, and C_d be the connected components of $G_\theta[R_d]$, identified by exact vertex sets. Additivity gives

$$V_\theta(d | B) = w(F) + w_d + \sum_{C \in C_d} \alpha_w(C), \quad (4)$$

where $\alpha_w(C)$ is the component’s MWIS value. For two actions in the same graph and boundary, set $\mathcal{A} = C_a \setminus C_b$ and $\mathcal{B} = C_b \setminus C_a$. Identical components cancel:

$$\begin{aligned} \Delta_{ab} &= V_\theta(a | B) - V_\theta(b | B) \\ &= w_a - w_b + \sum_{C \in \mathcal{A}} \alpha_w(C) - \sum_{C \in \mathcal{B}} \alpha_w(C). \end{aligned} \quad (5)$$

Sound component intervals $[L_C, U_C]$ imply

$$\begin{aligned} \Delta_{ab} &= w_a - w_b + \sum_{\mathcal{A}} L_C - \sum_{\mathcal{B}} U_C, \\ \bar{\Delta}_{ab} &= w_a - w_b + \sum_{\mathcal{A}} U_C - \sum_{\mathcal{B}} L_C. \end{aligned} \quad (6)$$

¹For a common satellite, equal starts or containment create conflicts. A crossing overlap $s_u < s_v < e_u < e_v$ conflicts exactly when $e_u - s_v < \tau_c$; nonoverlapping intervals conflict when $s_v - e_u < \tau_t$.

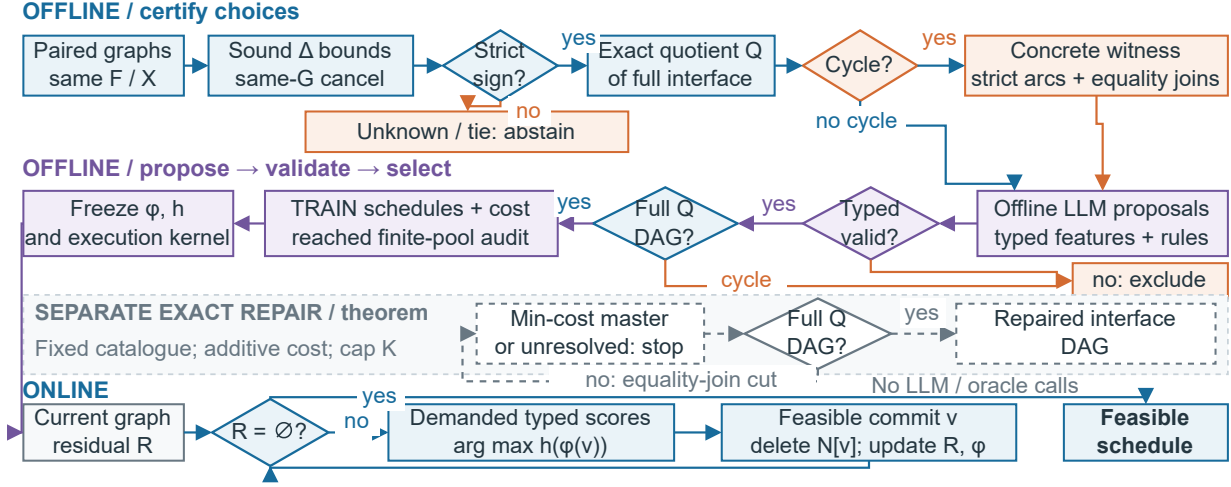


Figure 3: Executable control flow and its guarantees. Unknown comparisons abstain, invalid proposals fail, and the practical bank gate rebuilds the full quotient before TRAIN selection. The separate dashed finite-catalogue subroutine supports the additive-cost theorem; the practical proposal bank is not asserted to attain that optimum. The online loop enforces encoded pairwise feasibility.

Lemma (sound cancellation). These endpoints enclose Δ_{ab} . *Proof.* Independent component optima unite feasibly; shared terms cancel and signed unmatched bounds enclose the remainder. $\square$

Matching is configuration-specific (Figure 2). Checked packings give lower bounds. A sound U_H for $H \subseteq C$ and verified clique partition $\{K_j\}$ of $C \setminus H$ give

$$U_C = U_H + \sum_j \max_{v \in K_j} w_v. \quad (7)$$

Relaxing cross-region edges proves soundness. Exact arithmetic, outward rounding and stopped search retain enclosures; exhausted calls return $[0, w(C)]$. Distinct partitions need not dominate.

Aligned configurations retain the same feasible actions and boundary. For objective tolerance $\epsilon \geq 0$, reversal requires

$$\underline{\Delta}_{ab}^{\theta_0} > \epsilon, \quad \overline{\Delta}_{ab}^{\theta_1} < -\epsilon, \quad (8)$$

or exchanged roles; equal strict signs certify preservation. Other intervals supply no strict label. Certified occurrences $x = (G_\theta, R, v)$ require

$$h(\phi(x^+)) > h(\phi(x^-)). \quad (9)$$

4.2 Exact Quotients and Finite-Catalogue Repair

The finite quotient Q_ϕ identifies exactly equal feature vectors and adds arcs $[x^+] \rightarrow [x^-]$, including self-loops. Its nine base inputs are reward, duration, degree, neighbor-weight sum/maximum, compatible-weight sum, ground/satellite gaps and residual count. Diagnosis uses every declared field, excludes IDs/splits, and audits omitted resource scalars before attributing aliases to structure.

Proposition 1 (finite representability). An unrestricted deterministic pointwise scalar satisfies every observed strict requirement iff Q_ϕ is acyclic. *Proof.* A cycle demands a strict decrease back to the same score. Longest outgoing-path lengths on a finite DAG satisfy all arcs and scale to any positive scoring margin. $\square$

Cycle witnesses comprise arcs $p_i \rightarrow n_i$ and joins $n_i \sim_\phi p_{i+1}$. For finite typed catalogue $\mathcal{F}$, $d_{Wf} = 1$ if feature f separates a join of witness W . Positive standalone costs c_f define the master

$$\begin{aligned} \min_{z \in \{0,1\}^{|\mathcal{F}|}} \quad & \sum_f c_f z_f, \\ \text{s.t.} \quad & \sum_f d_{Wf} z_f \geq 1 \quad \text{for each retained } W, \\ & \sum_f z_f \leq K. \end{aligned} \quad (10)$$

Trials replace additions, retaining the base; surviving full-quotient cycles add cuts. Refined cycles can project to repeated coarse classes; rechecking one witness is insufficient.

Proposition 2 (finite-catalogue repair). Globally solved masters followed by an acyclic full quotient minimize additive cost for fixed catalogue, evidence and K . *Proof.* Necessary witness cuts give a lower bound on every valid repair; the final DAG attains the master bound. $\square$

Arbitrary finite tables encode weighted set cover: element self-loops are split by covering-subset features. Fixed K allows $O(|\mathcal{F}|^K)$ trials; variable K permits that reduction. Exhausted search or uncovered witnesses stay unresolved. This guarantee concerns catalogue repair; practical banks instead test each proposed interface for acyclicity. Neither test proves bounded-rule fit or transfer.

4.3 Typed Synthesis and Frozen Execution

LLM packets contain typed operations, graph/boundary examples, schedule feedback and optional certified relations/equality joins. It proposes a typed feature interface and deterministic scalar rule; tools enforce types, size and finite outputs (Figure 2). Packing/cover

summaries and edge statistics, such as

$$T_R(v) = \sum_{\{u,z\} \in E_\theta[N_R(v)]} \min(w_u, w_z), \quad (11)$$

are scorer inputs, not conditional-completion certificates. In Figure 1, neighborhood cover scoring gives $(a, b) : (8/26, 8/20) \rightarrow (8/14, 8/20)$; other actions can outrank both.

Shared expression DAGs and lazy inputs preserve defined finite reference scores and deterministic ties. For deleted $x \in N_R(v)$, sequential updates use

$$T(v) \leftarrow T(v) - \sum_{y \in N_R(v) \cap N_R(x)} \min(w_x, w_y), \quad (12)$$

using updated neighborhoods and exact accumulators. Nonincremental expressions use the snapshot interpreter. A reviewed local-dependency heap refreshes the neighbor frontier computed before deletion. Global or unreviewed dependencies, including untaken branches, require full rescore. Compilation, initialization, updates and queries remain charged. The kernel deploys the frozen program without LLM/oracle calls.

4.4 Action Audits and Joint Selection

At sampled reached boundaries, programs' argmax choices form finite pool $\mathcal{U}_B$. For chosen $d \in \mathcal{U}_B$,

$$\rho_{\mathcal{U}_B}(d) = \max_{u \in \mathcal{U}_B} [V_\theta(u | B) - V_\theta(d | B)]. \quad (13)$$

Maxima of zero and difference endpoints enclose this pool regret. Report coverage/unknowns separately from counterfactual states. Neither all-action regret nor complete-rollout loss is guaranteed.

TRAIN selection gates declared-interface acyclicity, then trades complete reward against measured work:

$$J(P) = q(P) - 0.002 [c(P) - 1]. \quad (14)$$

Here q, c average context reward/work ratios equally by family, against fixed feasible references and positive degree work. Failed runs contribute ratios 0, 100. Measured shared-DAG cost differs from additive catalogue cost. All pilot arms maximize J , break ties by original slot, and omit regret ties. Empty eligible banks return no program. Banks, evidence and selections freeze before fresh evaluation. Kernel feasibility does not guarantee quality.

5 Experiments

5.1 Populations, Baselines and Metrics

TRAIN contains 33 pairs/66 contexts at 32–128 contacts. Frozen confirmation uses 456 test contexts: 216 standard, 216 dense/long at 64–256 contacts, and 24 C3 source blocks at 64–512. Another 132 validation contexts do not select programs. Paired configurations preserve contacts, weights and windows while changing one resource parameter. Test ground gaps $(0.5, 6)$ shift from TRAIN $(0, 2)$; dense horizons shorten $1.8n \rightarrow 0.18n$ and maximum duration grows $12 \rightarrow 48$. Seeds/C3 opportunities exclude inventoried prior inputs.

Table 1 compares actually executed published implementations with frozen synthesis and classical controls. CHILS-ILS uses the published CHILS implementation's one-state ILS setting; it is not a separate algorithm. Public transfer retains 18 DIMACS originals

and 30 SATLIB instances, each with unit/hash weights. Independent decoders verify clique complements and literal-conflict reductions. These graph tasks do not encode satellite physics. Free and rule-only synthesis are project controls, not EoH/ReEvo reproductions.

Competitive reward divides each outcome by its strongest saved verified feasible witness, then averages contexts. Native seeds $1/2/3$ are averaged first; failed runs receive zero. Coverage uses all requested runs. A 100% ratio does not prove optimality; exact clique bounds and floating HiGHS endpoints remain separate. Native solvers receive five seconds; constructive policies use a cooperative five-process-CPU-second target, HiGHS ten seconds, and classical exchange two after the best of five fixed-rule initializers. Exchange elapsed time includes all five initialization attempts. Total computation is unmatched. Paired source-cluster 95% intervals use 2,000 bootstrap replicates, preserving configurations/shared seeds and C3 blocks; public families/weight modes stay separate.

5.2 What the LLM and Representation Contribute

The original 93-program TRAIN bank contains 12 new guided proposals, historical synthesis controls and fixed rules. One continuing assistant session supplies the new proposals; unequal authoring counts/compute preclude causal comparisons. TRAIN completes 5,996/6,138 assignments. Its primary normalizes weight by the larger of own weight and a deterministic neighbor clique-cover envelope. The regret tie-break and primary-only ablation select the identical AST.

The selected program improves Degree by 0.98 and 1.64 percentage points on standard/dense scheduling, with positive paired intervals. Against rule-only synthesis, gains are 0.54 [0.38, 0.69] and 1.59 [0.88, 2.30]. Free synthesis has higher standard/dense point means; their paired differences cross zero. C3 primary-minus-Free is +0.175 [0.036, 0.346], while the Degree and rule-only contrasts cross zero. Thus the evidence supports some benefits of this selected structural rule, without establishing an overall advantage attributable to LLM guidance. Published solvers and classical exchange attain higher fresh reward; several native implementations also terminate faster (Table 1). For example, Struction's median elapsed seconds are 0.009/0.015/0.012 on standard/dense/C3, versus primary's 0.209/0.235/0.474. This reports observed costs under different prescribed budgets, not equal-compute dominance.

Representation evidence differs in scope. Constructed probes certify obstructions; 200 physical-source alias queries remain unresolved. Pure TRAIN cancellation fixes unmatched bounds on 396 comparisons: all intervals nest, 199 narrow, and pooled median width falls $86.75 \rightarrow 36$. Table 2 exposes the mechanism's boundary. In standard TRAIN, 197 of 216 comparisons narrow, but none acquires a strict sign. Dense/long retains 17 strict certificates without narrowing; C3 has two narrower intervals and unchanged strict yield of 16. Removing common uncertainty exactly can leave the unmatched continuation's sign unresolved. A separate $10 \rightarrow 33$ strict-yield comparison changes the bounding strategy and does not isolate cancellation. Catalogue repair proves neither natural incidence nor unrestricted minimality.

Matched cold-authoring pilot. Four independent authoring blocks compare A (relations plus equality-join witnesses), B (relations)

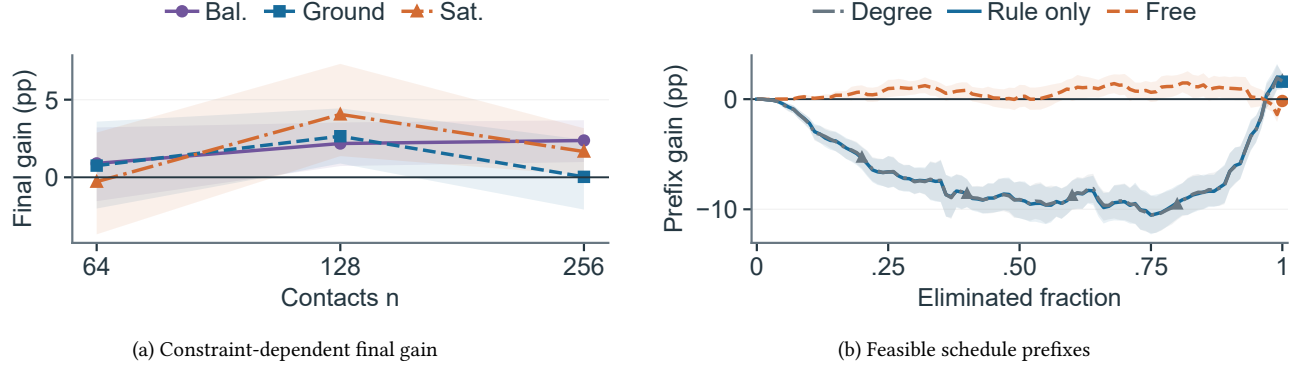


Figure 4: Quality by constraint regime and schedule progress. Left: primary-minus-rule-only gain on 216 dense/long contexts, with paired source-cluster intervals. Right: feasible-prefix gaps against three controls, with pointwise bands. Early deficits can recover by completion. Progress is eliminated-vertex fraction, not time or LLM learning.

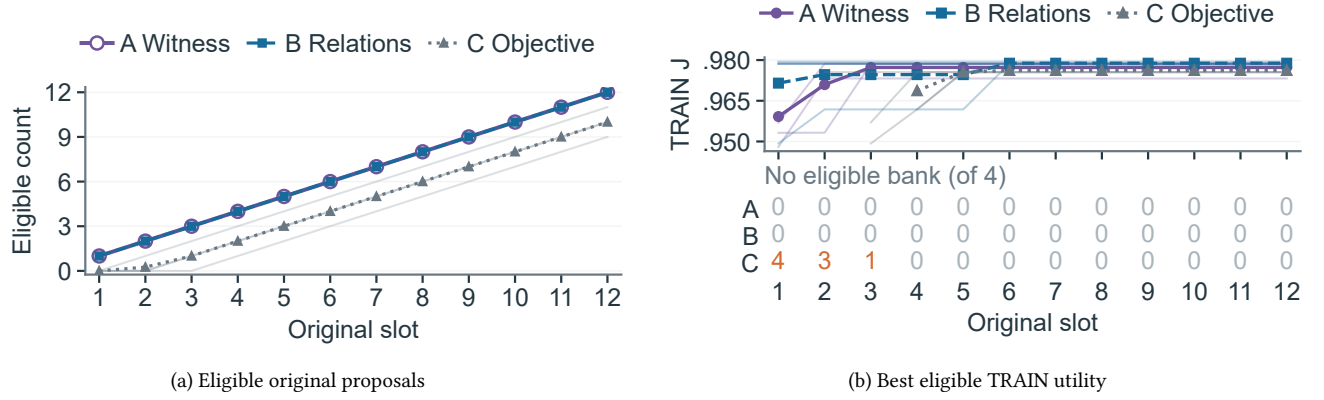


Figure 5: Does decision evidence help synthesis? Original-slot TRAIN curves: thin lines show four banks; thick means require every bank to have an eligible prefix. The strip counts missing prefixes. Full-interface DAG eligibility does not imply scalar rule fit. Witnesses add no final mean utility over labels alone; no TEST selection occurs.

Table 1: Executed baselines on seven separate populations. Cells: failure-zero competitive reward %, with completed-run % in parentheses. Native quality averages three seeds within each context; budgets are distinct. Blue identifies the study policy; bold marks the largest displayed quality per column, including ties, without significance claims.

Method	Standard 216	Dense/long 216	C3 24	DIMACS unit: 18	DIMACS hash: 18	SATLIB unit: 30	SATLIB hash: 30
<i>Competitive reward (run coverage), both %</i>							
CHILS [18]	100.00 (100)	100.00 (100)	100.00 (100)	99.63 (100)	99.94 (100)	99.97 (100)	100.00 (100)
CHILS-ILS [18]	100.00 (100)	100.00 (100)	100.00 (100)	100.00 (100)	99.81 (100)	99.91 (100)	99.93 (100)
M ² WIS [17]	100.00 (100)	100.00 (100)	100.00 (100)	88.89 (88.9)	100.00 (100)	99.71 (100)	99.99 (100)
Struction [15]	100.00 (100)	100.00 (100)	100.00 (100)	98.94 (100)	99.44 (100)	96.92 (100)	98.05 (100)
WeightedBR [21]	100.00 (100)	99.99 (100)	99.91 (100)	78.44 (88.9)	88.84 (96.3)	99.64 (100)	99.73 (100)
HiGHS MILP [19]	100.00 (100)	100.00 (100)	100.00 (100)	82.59 (100)	94.22 (100)	98.05 (100)	99.96 (100)
Degree	98.47 (100)	94.46 (100)	97.04 (100)	85.04 (94.4)	84.92 (94.4)	98.80 (100)	92.14 (100)
Rule-only synthesis	98.92 (100)	94.52 (100)	96.74 (100)	85.04 (94.4)	85.61 (94.4)	98.80 (100)	93.90 (100)
Free synthesis	99.53 (100)	96.27 (100)	97.39 (100)	82.20 (94.4)	84.10 (94.4)	32.71 (33.3)	29.67 (33.3)
Enumeration	98.90 (100)	94.50 (100)	96.75 (100)	85.80 (94.4)	85.46 (94.4)	32.86 (33.3)	31.18 (33.3)
Classical exchange	99.84 (100)	98.73 (100)	99.57 (100)	92.95 (100)	95.81 (100)	98.94 (100)	98.29 (100)
Guided primary	99.46 (100)	96.11 (100)	97.56 (100)	80.53 (94.4)	87.07 (94.4)	32.93 (33.3)	31.05 (33.3)

Table 2: Precision is not new preference information. Pure TRAIN cancellation fixes unmatched bounds. Widths are in original reward units; different weight scales preclude ranking family difficulty.

Family	Pairs	Narrower	Median width	Strict
Standard	216	197	100.5 → 23	0 → 0
Dense/long	108	0	43.875 → 43.875	17 → 17
C3	72	2	17060 → 17060	16 → 16

and C (objective feedback). All arms receive the same graphs, typed grammar, limits and 12 slots; tokens/latency are unmatched. The complete 144-slot bank freezes before TRAIN assessment. Each arm uses the same J selector, then evaluates its four frozen winners on 216 new standard/dense contexts, disjoint from inventoried prior inputs. Table 3 uses a common independently verified clique upper bound, not the competitive-reference denominator above.

All 2,808 TEST assignments complete. Relations improve four-block mean quality over objective feedback by 0.657 points, with positive differences in three blocks and one reversal. Explicit witnesses add only 0.014 points over relations (two ties). Figure 5 shows full-gate yields 48/48, 48/48 and 40/48; all actual-only quotients are acyclic, so the yield contrast arises from included prior diagnostic requirements. No candidate fits all 824 scalar rankings. This pilot supports a small descriptive label-guidance benefit, not strong witness superiority or a model-population effect; it contains no LLM-versus-non-LLM causal arm.

An additional fixed-program diagnostic samples at most eight equal-base competing edges per state, ordered before oracle queries. Thirteen complete unit-weight DIMACS graphs with at most 256 vertices yield 2 strict, 28 exact tie and 74 unresolved comparisons. Independently reconstructed induced subgraphs of at most 32 vertices from 48 public sources yield 65 strict and 155 exact ties; 164 unused query slots reflect absent eligible edges. All 67 strict comparisons are independently verified, including the four large conditional optima through a separate complement-coloring decision proof. Table 5 shows both complete-graph counterexamples. The primary’s added feature separates only one, and its scalar rule fits neither. These are public-graph information failures, not evidence of physical scheduling incidence or successful automatic repair.

5.3 Progress Curves and Transfer Limits

Figure 4 resolves aggregate gains by size and resource regime. Dense satellite-scarce $n = 128$ gains over rule-only are 4.05 [1.37, 7.31] points, whereas ground-scarce $n = 256$ gives 0.03 [−2.08, 2.42]. Actual rollout-prefix curves lag Degree/rule-only early and recover a positive final mean; the Free contrast remains inconclusive. Prefixes are indexed by eliminated-vertex fraction, not elapsed time or LLM learning iterations. These retrospective diagnostics replay unchanged frozen ASTs and do not supply unrecorded native anytime trajectories.

Primary completion is 456/456 fresh contexts, 34/36 DIMACS and 20/60 SATLIB; public quality losses include timeout failures rather than selecting successful graphs. CHLS, ILS and Struction complete all 288 public short seed runs; M²WIS/WeightedBR fail six/eight. Separate prescribed 30-second subsets have 20 fresh and

21 public contexts; native methods complete all runs, while public primary completes 18/21 at its unchanged CPU target. Four SNAP simple projections (eight weight variants, 4,039–12,008 vertices) are a separate extension: original primary completion is 0/8. The SATLIB failures are concentrated: all ten completed cases per weight mode are uf20–91, while all uf50–218/uf75–325 cases time out. Successful-subset competitive reward is 98.79%/93.14%, but its failure-zero population means are 32.93%/31.05%. Conditional quality would hide a source-family selection effect; Degree completes all thirty cases. On DIMACS unit weights, primary also trails Degree among completed cases, so computation alone does not explain transfer loss. Completed C3 schedules pass the original physical verifier. A fixed-AST 588-context audit finds primary finite-pool regret positive/zero/unknown in 18/191/346 actually reached sampled test states; this is not all-action or whole-schedule regret.

5.4 Execution Adaptation

Demanded/full-interface traces match on all 456 fresh contexts. A separate same-AST heap extension preserves all 2,736 assessable fresh pairs; mean context-median full-scan/heap CPU ratios are $15.45\times$ [12.91, 18.19], $1.19\times$ [1.15, 1.23] and $4.61\times$ [4.09, 5.19] on standard/dense/C3. Degree slows on dense (0.87 $\times$). Sparse full scans complete 0/24 per AST; five-second heaps complete 7/24 primary and 18/24 Degree, increasing separately to 12/24 and 24/24 at thirty seconds. With no successful full-scan pairs, sparse speedup and backend parity remain unassessable. These execution gains do not measure LLM benefit or replace failed original confirmations (Figure 6).

Frozen-winner transfer. An exploratory extension executes the 12 cold-authoring winners and Degree on 120 earlier public/C3 contexts, retaining prior verified upper bounds and no reselection. All use the same compiled backend and cooperative five-CPU-second target after AST parsing. Table 4 reports 1,560 assignments: 1,007 complete and 553 time out. Witness guidance has no general transfer advantage. All synthesized arms complete SATLIB at 273 vertices and fail at 654/975; Degree also completes 654. The Witness block with extra neighbor packing has five C3 timeouts. Objective’s C3 quality exceeds compiled Degree by 0.758 points of U , but reaches only 98.12% of the prior feasible reference. Reused contexts make this a diagnostic extension, not independent confirmation.

6 Discussion and Limits

Designed probes certify temporal contradictions; 200 physical-source alias queries remain unresolved. Public counterexamples demonstrate separate input failures, without physical incidence or successful repair. The acyclic primary interface still fits only 631/824 TRAIN rankings (788 actual-action, 36 prior): finite representability does not guarantee bounded-rule fit or transfer.

Repair optimality covers fixed evidence, a finite catalogue and additive standalone costs. Shared cost and quality remain empirical. Finite-pool regret excludes unqueried actions; cancellation narrows uncertainty without new strict signs.

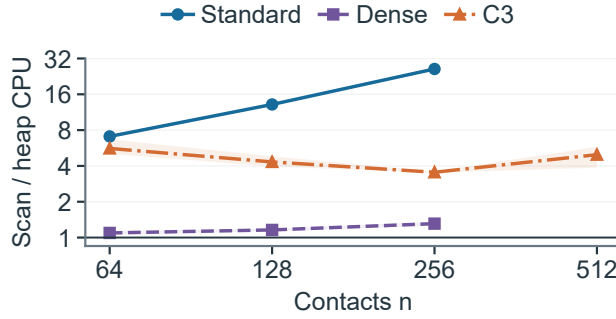
The four-block pilot has unmatched tokens/latency, small label gains and little witness gain; exploratory transfer adds no general advantage. Earlier authoring counts differ. Fresh confirmation excludes prior instances, but C3 has one physical source. Hash

Table 3: Four-block matched pilot on 216 fresh graphs. Cells: failure-zero mean $100W/U$, with verified clique upper U . All original slots are retained. Contrasts are percentage points. Blue highlights Witness and the relations-objective contrast; four descriptive blocks establish no significance.

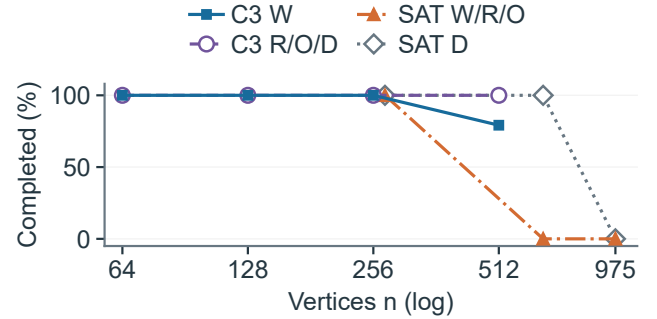
Arm	Block 1	Block 2	Block 3	Block 4	Eligible/48	Complete/864
A Witness	79.38	79.39	79.38	79.38	48/48	864/864
B Relations	79.38	79.38	79.33	79.38	48/48	864/864
C Objective	78.48	78.48	79.38	78.48	40/48	864/864
Degree	78.57 (one common reference)				–	216/216
A–B	+0.00	+0.01	+0.05	+0.00	mean +0.01 pp	
B–C	+0.89	+0.89	-0.05	+0.89	mean +0.66 pp	

Table 4: Exploratory transfer on 120 reused contexts, without reselection. Cells: failure-zero mean $100W/U$ (completed/assigned), using prior clique upper U and equal authoring-block weights. All arms use the compiled five-CPU-second target after AST parsing. Blue identifies Witness; bold marks displayed column maxima, including ties, without significance.

Arm	DIMACS unit	DIMACS hash	SATLIB unit	SATLIB hash	C3
A Witness	40.15 (63/72)	41.28 (65/72)	24.65 (40/120)	22.60 (40/120)	65.60 (91/96)
B Relations	40.11 (64/72)	40.82 (64/72)	24.65 (40/120)	22.61 (40/120)	68.63 (96/96)
C Objective	39.85 (63/72)	42.23 (67/72)	24.65 (40/120)	23.42 (40/120)	68.99 (96/96)
Degree (compiled)	43.94 (17/18)	41.72 (17/18)	49.49 (20/30)	44.44 (20/30)	68.23 (24/24)



(a) Same-program execution cost



(b) Frozen-winner transfer completion

Figure 6: Execution gain and its remaining limits. Left: mean context-median scan/heap CPU ratios on 456 contexts, three order-balanced repeats, with source-cluster intervals; all 1,368 pairs match. C3 size groups have three clusters. Right: exploratory completion under the same compiled five-CPU-second target, at measured sizes only. Exactly equal rate curves share a line; SATLIB weights pool only completion. The panels use separate protocols and measure neither LLM benefit nor reward gain.

Table 5: Certified complete-public-graph counterexamples. Roots have equal nine-input vectors and unit reward. V includes the forced root; n_c is the frozen primary’s neighbor-cover feature.

Graph	$V(a)$	$V(b)$	$n_c(a)$	$n_c(b)$	Rule
brock200_2	10	9	23	23	Tie
san200_0.7_1	30	25	18	17	Wrong

weights/simple projections are declared extensions. Encoded feasibility implies neither greedy optimality nor published-solver dominance.

7 Conclusion

Certified conditional preferences expose lost distinctions through exact quotient contradictions. Equality-join witnesses direct typed feature synthesis; full-quotient separation selects finite-catalogue repairs. Complete schedules and compiled work guide feature-rule selection. The frozen program adapts within a feasibility-enforcing kernel without online LLM or oracle calls, making information adequacy explicit while separating its guarantees from measured performance.

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